Andy **Doan**

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Education

University of Sydney
Sydney, Australia

Master of Computer Science | Advanced entry | WAM: 82.3 — *Feb. 2025 - Nov. 2026 (Expected)*

SP Jain School of Global Management
Sydney, Australia

Bachelor of Data Science | Dean's List | CGPA: 3.5 — *Sep. 2021 - May. 2024*

Work Experience

Digital Nexus AI, Software Engineering Intern — *Melbourne, Australia (Remote), Dec. 2025 - Apr. 2026*

- Built a **shared LLM module** that wraps OpenAI, Anthropic, and Gemini behind one interface, letting other teams build an AI feature in ∼**10 lines instead of** ∼**100** across **20+** internal use cases.
- Built a **document similarity pipeline** using **Winnowing fingerprinting** to flag near-duplicate submissions, feeding analysis and AI plagiarism-detection features.
- Built a **real-time WebSocket service** supporting **500+** concurrent users that flags inactive students, alerting tutors and nudging the students during live tutorial sessions.
- **Technologies**: Python, TypeScript, FastAPI, Express.js, PostgreSQL, Pytest, Jest.

VPBank, Software Engineer — *Hanoi, Vietnam, Jul. 2024 - Feb. 2025*

- Designed a **serverless metadata ingestion pipeline** using AWS Glue to scan **5K+** tables weekly, landing structured schema data in S3, replacing manual schema collection and feeding downstream catalog generation.
- Built an **automated metadata generation service** using AWS Glue & Bedrock to catalog **5K+** tables in OpenMetadata, achieving ∼**90% accuracy**, enabling staff to discover bank data without manual documentation.
- Deployed a **production chatbot** on Amazon ECS (Linux containers) giving **200+** staff on-demand knowledge of bank data and regulations, achieving ∼**90%** answer accuracy and ∼**30%** faster response time through optimized indexing.
- Engineered a **cost monitoring service** tracking **$10K+/month** in AWS foundation model usage across **20+** teams, offloading usage tracking to async workers on a **Redis** queue with retries to keep the API responsive under load.
- Built **10+ production ETL pipelines** using IBM DataStage, processing **millions of records weekly** across business units, supporting downstream analytics and reporting.
- **Technologies**: Python, FastAPI, PostgreSQL, Redis, IBM DataStage, AWS (S3, Glue, Bedrock, ECS, CloudWatch).

Projects

AstroDreamers | React.js, Spring Boot, FastAPI, PostgreSQL | Demo | GitHub — *2025 NASA Space Apps Challenge*

- Built an **award-winning air-quality platform** on NASA TEMPO/OpenAQ APIs with **85% forecast accuracy**.

MicroCart | React.js, Spring Boot, RabbitMQ, PostgreSQL | GitHub | Article

- Built an **e-commerce order backend** as 4 microservices that coordinate over **RabbitMQ** message queues.
- **Saga pattern:** rolled back failed transactions across services with compensating actions, keeping data consistent.

Remote Canvas Server | C, POSIX Threads, IPC | GitHub

- Built a **terminal-based animation server** in C that renders shapes and sprites from multiple clients into MP4 video.
- **Concurrent processing:** worker threads run commands from a shared queue in parallel, returning replies in order.

Honors & Awards

- **Team Lead**, NASA Space Apps Challenge | Sydney Local Winner (Top 2/32 teams). *2025*
- **Top 18/132 Finalist**, VPBank Generative AI Hackathon | Selected for VPBank Talent Program. *2024*
- **1st Place**, SP Jain DSA Competition (Top 1/51 participants). *2022*

Skills

- **Languages & Frameworks**: Go, Java, Python, C, TypeScript, SQL, Spring Boot, FastAPI, Express.js, React.js.
- **Cloud, DevOps & Databases**: AWS (S3, Glue, Bedrock, ECS), PostgreSQL, Redis, RabbitMQ, Docker, Linux, CI/CD.